

IT5437 Assignment 2 on Fitting and Alignment

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1. Table 1 shows the first five lines of the dataset in the file `lines.csv`. It represents three points scatters conforming to three lines. The coordinates of the points are stored in columns x_1 , x_2 , x_3 , y_1 , y_2 , and y_3 .
 - (a) Use total least squares to fit the lines to only using the data corresponding to the first line. Report the resulting parameters.
 - (b) Now, use all the points as indicate in the code snippet below and fit three lines. Hint: Run RANSAC to find a line, mask the consensus and run again and so on to find the three lines

```
D = np.genfromtxt("lines.csv", delimiter=",", skip_header=1)
X_cols = D[:, :3]
Y_cols = D[:, 3:]
X_all = X_cols.flatten()
Y_all = Y_cols.flatten()
```

Table 1: First five rows of the dataset

x_1	x_2	x_3	y_1	y_2	y_3
-5.3055	-4.0601	-5.2613	-12.6663	-3.7962	3.6917
-5.5404	-5.0032	-3.9926	-11.0077	-3.9856	4.9000
-4.9821	-4.5845	-4.3312	-11.6973	-3.5893	5.0469
-4.4957	-5.0641	-4.7820	-11.9780	-3.5971	4.6359
-4.4422	-4.4114	-4.5675	-12.4150	-2.7995	4.7397

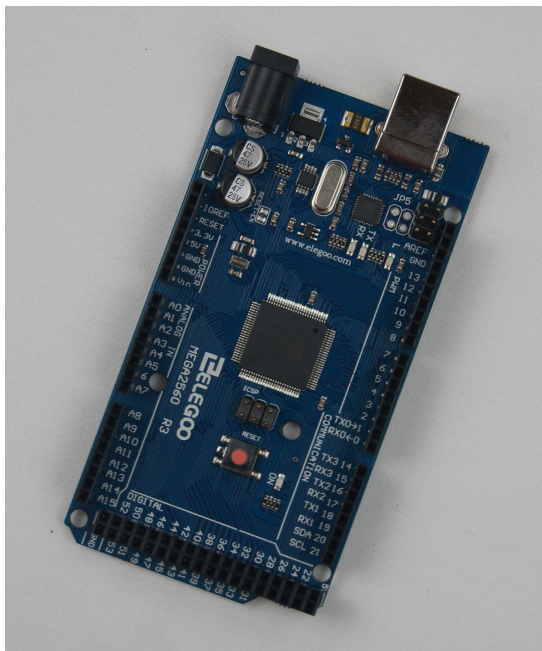
2. Fig. 1 shows an image of a pair of earrings. Assume that a camera is mounted with the optical axis perpendicular to the imaging plane (where the flat earrings are kept). If the focal length of the camera is 8 mm, the pixel size is $2.2 \mu\text{m} \times 2.2 \mu\text{m}$, and the distance from the lens to the imaging plane is 720 mm, what are the sizes of the earrings?
3. Fig. 2 two circuit boards¹. Listing 1 shows a code snippet that can be used to click and obtain the four corners. It is possible to transform the image of one circuit board to the perspective of the other using the homography transformation. This can be used to find the differences between the two circuit boards. Perform the following steps:
 - (a) Manually click about 6 corresponding points in each circuit board, compute the homography, and warp one image to the perspective of the other. Display the warped image.
 - (b) Subtract the warped image from the other image to find the differences between the two circuit boards. Display the difference image.

¹In fact, the two images are of the same circuit board, manually transformed.

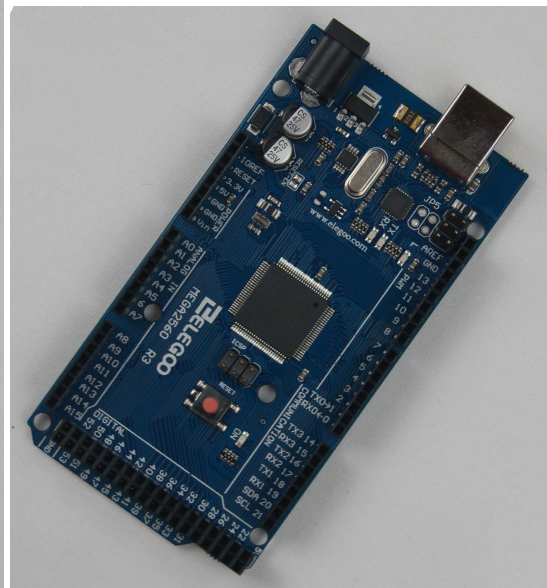


Figure 1: Earrings.

- (c) Use SIFT or ORB to find keypoints and descriptors in both images, match the descriptors, and display the matches. You can use OpenCV functions for this purpose.
- (d) Use the matches to compute a homography and warp one image to the perspective of the other. Display the warped image and the difference image as in steps (a) and (b). Compare the results with those obtained in steps (a) and (b). Discuss any differences in the results and possible reasons for those differences.



(a) Circuit board c_1 .



(b) Circuit board c_2 .

Figure 2: Two circuit boards.

Listing 1: Code for clicking a quadrangle.

```
import cv2 as cv
import numpy as np

N = 5
global n
n = 0
p1 = np.empty((N,2))
```

```

p2 = np.empty((N,2))

DISPLAY_W, DISPLAY_H = 900, 650

# mouse callback function
def draw_circle(event,x,y,flags,param):
    global n
    p = param[0]
    if event == cv.EVENT_LBUTTONDOWN:
        cv.circle(param[1],(x,y),5,(255,0,0),-1)
        p[n] = (x,y)
        n += 1

im1 = cv.imread('a2_images/c1.jpg', cv.IMREAD_REDUCED_COLOR_4)
im2 = cv.imread('a2_images/c2.jpg', cv.IMREAD_REDUCED_COLOR_4)
im1copy = im1.copy()
im2copy = im2.copy()

cv.namedWindow('Image 1', cv.WINDOW_AUTOSIZE)

param = [p1, im1copy]
cv.setMouseCallback('Image 1',draw_circle, param)

while(1):
    cv.imshow("Image 1", im1copy)
    if n == N:
        break
    if cv.waitKey(20) & 0xFF == 27:
        break

param = [p2, im2copy]
n = 0
cv.namedWindow("Image 2", cv.WINDOW_AUTOSIZE)
cv.setMouseCallback('Image 2',draw_circle, param)

while(1):
    cv.imshow("Image 2", im2copy)
    if n == N:
        break
    if cv.waitKey(20) & 0xFF == 27:
        break

print(p1)
print(p2)

```

GitHub Profile

You must include the link to your GitHub (or some other SVN) profile, so that I can see that you have worked on this assignment over a reasonable duration. Therefore, make commits regularly. However, I will use only the pdf for grading to save time.

Submission

Upload a report (four pages or less) named as your_index_a02.pdf. Include the index number and the name *within the pdf* as well. The report must include important parts of code, image results,

and comparison of results. The interpretation of results and the discussion are important in the report. Extra-page penalty is 20 marks per page.